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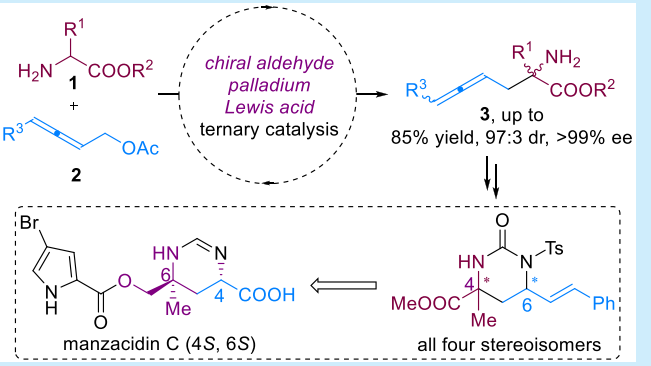
Core Structure-Oriented Asymmetric α-Allenylic Alkylation of Amino Acid Esters Enabled by Chiral Aldehyde/Palladium Catalysis

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ABSTRACT: Aiming at the reported chiral synthons leading to manzacidins A and D, here we report a highly efficient catalytic asymmetric α-allenylic alkylation reaction of NH_2-unprotected amino acid esters that is promoted by combined chiral aldehyde/palladium catalysis. Fifty examples of unnatural α,α-disubstituted amino acid esters are reported with good-to-excellent yields and stereoselectivities. Based on this methodology, a key intermediate leading to manzacidin C and its other three stereoisomers is prepared accordingly.



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Manzacidins are important members of the bromopyrrole alkaloid family.[1] Because of their broad biological activities and unique tetrahydropyrimidine cores, they have attracted significant attention from synthetic chemists.[2] The tetrahydropyrimidine cores of manzacidins feature chiral 1,3-tertiary-quaternary carbon centers and C-6 epimers. Therefore, the precise construction of these chiral units is an essential prerequisite to the preparation of the corresponding natural products. In 2000, Ohfune and co-workers reported the first total synthesis of manzacidins A and C.[3] Subsequently, various elegant asymmetric methods—including the use of chiral synthons,[4] chiral auxiliaries,[5] and chiral catalysts[6]—have been reported for the preparation of manzacidins and their analogues. However, the majority of reported methods rely on the use of stoichiometric amounts of chiral reagents with few catalytic asymmetric routes. Therefore, highly efficient catalytic asymmetric methodologies for the construction of the corresponding chiral tetrahydropyrimidine core are required.

In Ohfune's work, chiral tetrahydropyrimidine **I** and its precursor tetrahydropyrimidine-2-one **II** are two intermediates that lead to manzacidin C.[3] Intermediate **II** was prepared from the chiral synthon (*S*)-2-aminopent-4-en-1-ol in seven steps. We envisaged that the carboxyl group of **II** could be prepared by oxidizing the corresponding alkene double bonds of **III** and that the chiral tetrahydropyrimidine-2-one core of **III** could be constructed by the asymmetric intramolecular addition of the NH group to the allene unit of compound **IV**.[7] Chiral urea−allene **IV** could be readily prepared by the addition of amino acid ester **V** to 4-methylbenzenesulfonyl isocyanate.[8] Thus, allene-containing chiral amino acid ester **V** was anticipated to be the key starting material for the preparation of manzacidin C and its family members. We envisaged that this amino acid ester could be prepared by the catalytic asymmetric α-allenylic alkylation of amino acid ester derivatives (Figure 1a).

The catalytic asymmetric α-allenylic alkylation of amino acid ester derivatives is an important strategy for the preparation of chiral amino acids[9] or chiral allenes.[10] Especially, Zhang, Wang, and several other research groups reported their elegant methods for constructing chiral amino acid molecules bearing both carbon and axial chiralities.[11] However, all reported methods employ N-protected amino acid esters as reactants. Chiral aldehyde catalysis can be used to promote the direct α-functionalization of N-unprotected amino acid derivatives.[12] In our previous work, we explored both chiral aldehyde-mediated organocatalysis[13] and combined organo/transition-metal catalysis.[14] In particular, we reported a chiral aldehyde-involved ternary catalysis that promoted the asymmetric α-allylation[15] and α-benzylation[16] of NH_2-unprotected amino acid derivatives. With the utilization of this reported reaction model,[15] we reasoned that the straightforward synthesis of the target allene-contained chiral amino acid ester **3** from the racemic reactants **1** and **2** with high efficiency via a dynamic kinetic asymmetric transformation (DyKAT) process[11a] could be achieved. Furthermore, the absolute configuration of the chiral quaternary carbon center might be easily switched by changing the chiral configuration of the catalyst, thereby providing an

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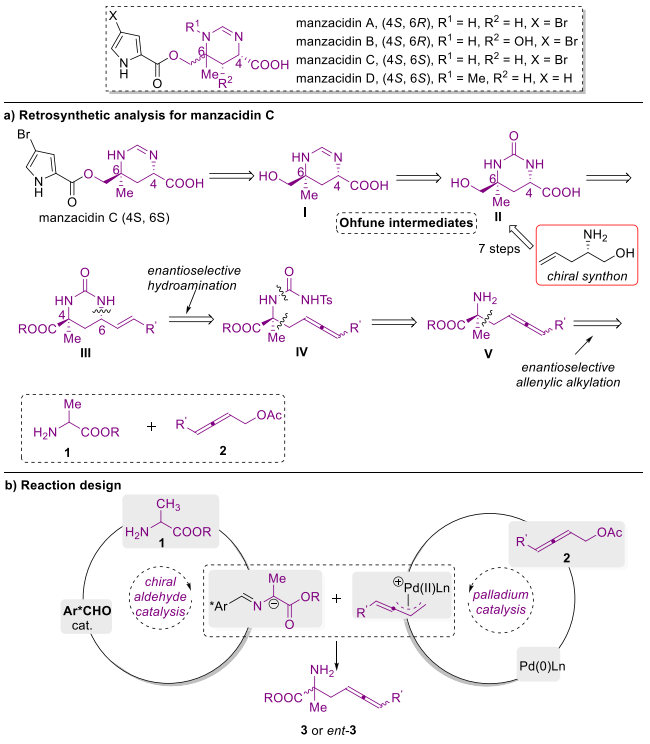


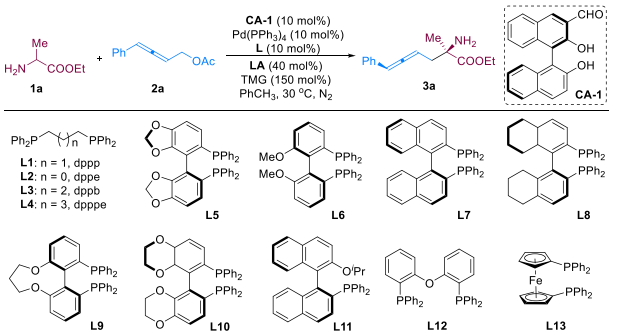
Figure 1. Core structure-oriented reaction design.

efficient route to the preparation of all stereoisomers of manzacidins A and D (Figure 1b).

With the above research plan, we chose ethyl alaninate **1a** and allenylic alkylation reagent **2a** as model reactants. The catalytic system combined chiral aldehyde **CA-1**, palladium ($[Pd(C_3H_5)Cl]_2$), and a Lewis acid ($ZnCl_2$). As expected, target product **3a** was generated in 20% yield, 90:10 diastereomeric ratio (dr), and 79% enantiomeric excess (ee) (Table 1, entry 1). The results improved with the use of $Pd(PPh_3)_4$ as the palladium source (Table 1, entry 2). Several achiral phosphine ligands were then tested; however, none of these ligands improved the experimental outcomes (Table 1, entries 4−7). Chiral phosphine ligands were investigated next (Table 1, entries 8−16). Notably, chiral ligands **L5** and **L6** significantly enhanced the stereoselectivities; however, the yields were still unsatisfactory (Table 1, entries 8 and 9). Increasing the equivalents of amino acid ester **1a** increased the yield of **3a** to 59% (Table 1, entry 17). By employing $ZnBr_2$ as the Lewis acid, the yield of **3a** was further increased (Table 1, entry 18). Due to its excellent stereoselective control, chiral ligand **L6** was reinvestigated. Using ligand **L6**, the yield, dr, and ee all improved relative to catalyst **L5** under equivalent conditions (Table 1, entry 22). Next, the reaction temperature and the equivalents of Lewis acid were tuned (see the Supporting Information). When this reaction proceeded at 15 °C and with 60 mol % Lewis acid $ZnBr_2$, product **3a** was generated in 83% yield, 95:5 dr, and >99% ee (Table 1, entry 24). A comparison experiment indicated that the *R*-configuration of chiral aldehyde **CA-1** matched with the *S*-configuration of chiral phosphine ligand **L6** (Table 1, entries 24 and 25). Based on these results, the reaction conditions listed in Table 1, entry 24, were chosen as the optimal conditions for subsequent investigations.

With the optimal reaction conditions established, we set out to complete the formal synthesis of manzacidin C and

Table 1. Optimization of Reaction Conditions[a]



entry	L	LA	yield (%)[b]	dr[c]	ee (%)[c]
1[d]	**L1**	$ZnCl_2$	20	90:10	79
2	**L1**	$ZnCl_2$	28	94:6	84
3[e]	**L1**	$ZnCl_2$	trace	-	-
4	**L2**	$ZnCl_2$	NR	-	-
5	**L3**	$ZnCl_2$	trace	-	-
6	**L4**	$ZnCl_2$	NR	-	-
7	PPh_3	$ZnCl_2$	NR	-	-
8	**L5**	$ZnCl_2$	34	93:7	95
9	**L6**	$ZnCl_2$	33	95:5	99
10	**L7**	$ZnCl_2$	trace	-	-
11	**L8**	$ZnCl_2$	14	85:15	92
12	**L9**	$ZnCl_2$	trace	-	-
13	**L10**	$ZnCl_2$	24	83:17	95
14	**L11**	$ZnCl_2$	trace	-	-
15	**L12**	$ZnCl_2$	16	68:32	76
16	**L13**	$ZnCl_2$	trace	-	-
17	**L5**	$ZnCl_2$	59	94:6	99
18	**L5**	$ZnBr_2$	67	88:12	98
19	**L5**	ZnF_2	12	90:10	98
20	**L5**	$Zn(OTf)_2$	14	79:21	88
21	**L5**	$Zn(OAc)_2$	20	91:9	98
22[f]	**L6**	$ZnBr_2$	71	94:6	98
23[f,g]	**L6**	$ZnBr_2$	78	93:7	99
24[f,g,h]	**L6**	$ZnBr_2$	83	95:5	>99
25	***ent*-L6**	$ZnBr_2$	61	66:34	92

[a]**1a** (0.45 mmol), **2a** (0.2 mmol), **CA-1** (10 mol %), **L** (10 mol %), [Pd] (10 mol %), LA (40 mol %), and TMG (150 mol %) in toluene (1.5 mL) at 30 °C. [b]Isolated yield. [c]The dr and ee values were measured with **3a**-Ts by chiral HPLC analysis. [d]With $[Pd(C_3H_5)Cl]_2$ (5 mol %). [e]With $Pd_2(dba)_3$ (10 mol %). [f]Toluene (1.0 mL). [g]At 15 °C. [h]With $ZnBr_2$ (60 mol %). NR = no reaction.

construct core tetrahydropyrimidine-2-one **II** with various chiral configurations. The synthetic route to manzacidin C is shown in Figure 2a. Under the optimal reaction conditions, chiral aldehyde *ent*-**CA-1** and ligand *ent*-**L6** catalyzed the formation of (*S*,*R*)-**3a** in 83% yield, 95:5 dr, and 99% ee. After (*S*,*R*)-**3a** was reacted with 4-methylbenzenesulfonyl isocyanate, product **4** was obtained in 90% yield. Next, the intramolecular addition of the amino group to the allene unit was studied. After various reaction conditions were screened (see the Supporting Information), rhodium catalysis was found to give the product, (*S*,*S*)-**5**, in 79% yield and 55:45 dr. This diastereoisomeric mixture was treated with magnesium under sonication, leading to (*S*,*S*)-**6** in 79% yield. The major diastereoisomer of (*S*,*S*)-**6** was isolated by silica column chromatography and submitted to subsequent oxidation/reduction reactions, affording (*S*,*S*)-**7** in 82% yield. Finally, the hydrolysis of (*S*,*S*)-**7** with LiOH gave the key chiral

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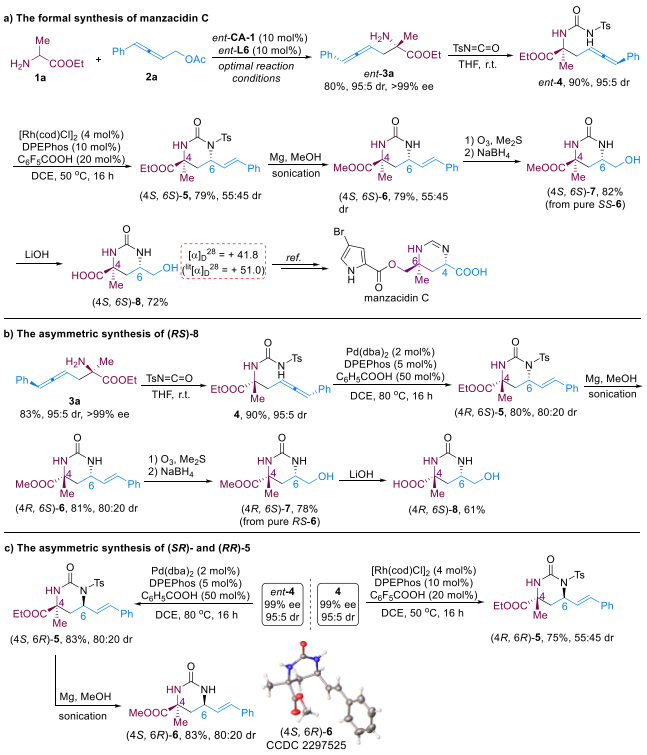


Figure 2. Formal synthesis of manzacidin C and stereodivergent synthesis of chiral tetrahydropyrimidine-2-one derivatives.

Ohfune intermediate (*S*,*S*)-**8** in 72% yield. The analytic data of (4*S*,6*S*)-**8** was identical to that reported in the literature, and the absolute configuration was assigned by comparing its optical rotation with literature values (see the Supporting Information).[3] Thus, the catalytic asymmetric formal synthesis of manzacidin C was achieved by using readily available and inexpensive starting materials.

Next, the synthetic route for the preparation of (4*R*,6*S*)-**8** was studied. The *R* configuration of the chiral quaternary carbon center was successfully installed by using chiral aldehyde **CA-1** and chiral phosphine ligand **L6** (Figures 2b and 3a). In the preparation of chiral Ohfune intermediate (*S*,*S*)-**8**, the *S*-configuration of the quaternary carbon center induced the *S*-configuration of the chiral C-6 carbon center under rhodium catalysis. Thus, we needed to find an alternative reaction condition to construct the chiral C-6 carbon center with an *S*-configuration under induction of the *R* quaternary carbon center. Fortunately, we found that palladium catalysis could promote the generation of (4*R*,6*S*)-**5** from compound **4**. Under the same manipulations used for the synthesis of (4*S*,6*S*)-**8**, (4*R*,6*S*)-**8** was successfully prepared (Figure 2b).

Considering the different stereoselectivities of the rhodium and palladium catalysts in the intramolecular hydroamination reaction, we reasoned that the preparation of another two isomers of compound **5** could be achieved by switching the starting materials and catalytic systems. As expected, (4*S*,6*R*)-**5** was generated from reactant *ent*-**4** using palladium catalysis, and the major component (4*R*,6*R*)-**5** was generated from **4** under rhodium catalysis (Figure 2c). The absolute configuration of (4*S*,6*R*)-**6** was established by single crystal X-ray diffraction analysis, and the absolute configurations of the other isomers of **5** were deduced from their ^{1}H NMR data and the original chiral induction sources. Thus, all four stereoisomers of core tetrahydropyrimidine-2-one **5** were successfully prepared using our catalytic asymmetric methodology.

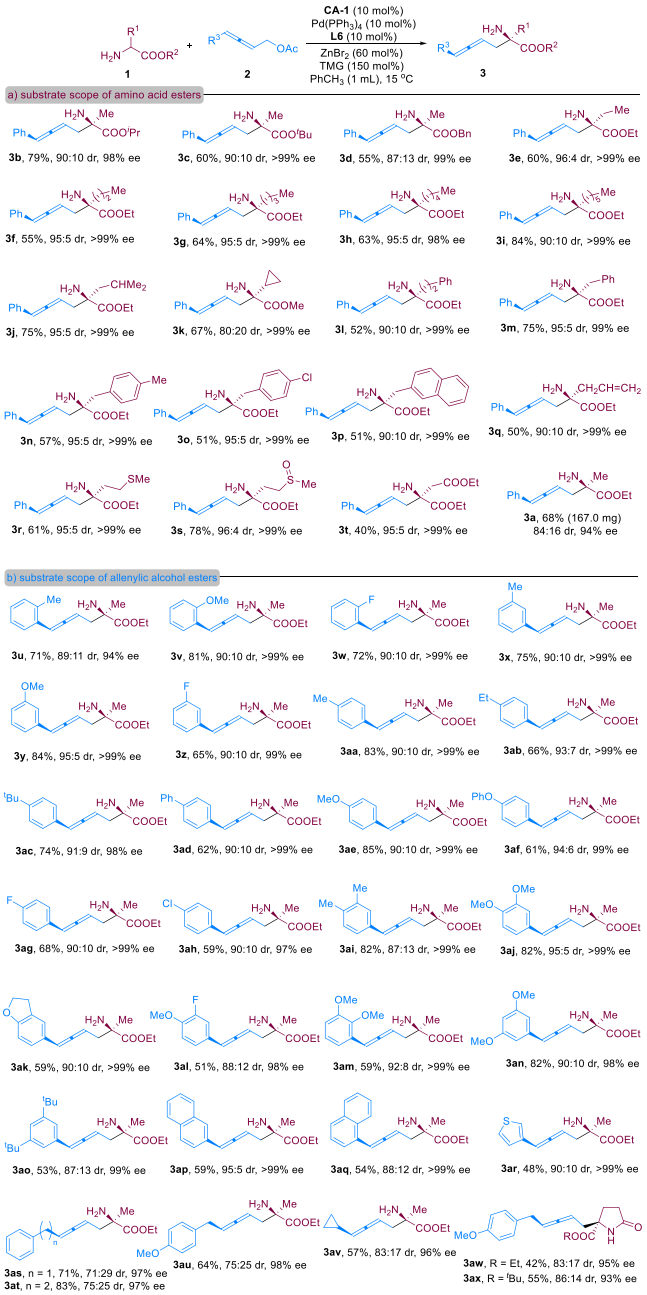


Figure 3. Substrate scopes.

Next, the application of these optimal reaction conditions to other substituted amino acid esters and allenyl alcohol esters was studied. First, various readily available amino acid derivatives were employed as donors in the reaction. Amino acid esters bearing different alkoxyl groups were well-tolerated in the reaction, affording the corresponding products in good yields, good diastereoselectivities, and excellent enantioselectivities (Figure 3a, **3b** and **3c**). These results indicated that the target products could be converted to the corresponding amino acids under various hydrolysis conditions. Linear and branched alkyl-substituted amino acid esters were then tested; all of these substrates afforded products in good-to-high yields and excellent stereoselectivities (Figure 3a, **3e**−**3k**). The phenylalanine ester and its analogues were also suitable

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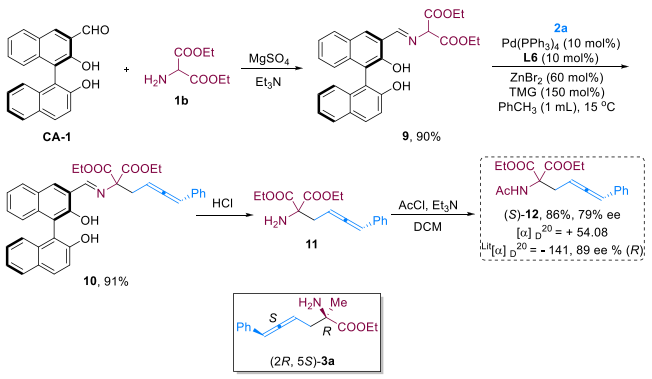
reaction partners for allenylic alcohol ester **2a**, leading to products **3l**−**3p** with good experimental outcomes. Other functional groups, such as an alkene, a thioether, a sulfoxide, and a carboxylic ester, were also tolerated under the optimal reaction conditions, giving the desired products **3q**−**3t** in moderate-to-high yields and excellent stereoselectivities. Furthermore, we found the model reaction could be carried out in a 1.0 mmol scale, leading to product **3a** in 68% yield (167.0 mg), 84:16 dr, and 94% ee (Figure 3, **3a**).

The substrate scope of the allenyl alcohol esters was also investigated. Phenyl-substituted allenylic alcohol esters bearing a single substituent on the benzene ring were good reaction partners for **1a**. The substitution positions and the electronic and steric effects of the substituent did not affect the experimental outcomes. Products **3u**−**3ah** were generated in good-to-high yields with excellent diastereoselectivities and enantioselectivities. Similar results were obtained when phenyl-substituted allenylic alcohol esters bearing two substituents were used (Figure 3b, **3ai**−**3ao**). Nonphenyl aryl-substituted allenylic alkylation substrates, such as naphthyl- and thionyl-substituted ones, also reacted smoothly with **1a** under the optimal reaction conditions, giving products **3ap**−**3ar** in satisfactory yields and excellent stereoselectivities. Several alkyl-substituted allenylic alcohol esters were then tested, all of which afforded the corresponding products in high yields, good diastereoselectivities, and excellent enantioselectivities (Figure 3b, **3as**−**3ax**). When a glutamate diester was used, allenylic alkylation and lactamization took place successively, leading to the corresponding optically active γ-lactam derivatives with good experimental outcomes (Figure 3b, **3aw** and **3ax**).

Under the influence of chiral aldehyde **CA-1** and chiral ligand **L6**, the Schiff base **9** reacted with **2a** and gave product **10** in 91% yield. After hydrolysis and protection of the amino group by acetyl, compound **12** was prepared. The absolute configuration of the chiral allene unit of molecule **12** was assigned as *S* by comparing its optical rotation value with the literature's data.[10a] Combining with the quaternary *S* configuration determined in (4*S*,6*R*)-**6**, the absolute configurations of the product **3a** were assigned as (2*R*,5*S*) (Scheme 1). The absolute configurations of the other products were assigned accordingly.

In conclusion, we rationally designed an asymmetric α-allenylic alkylation reaction of NH_2-unprotected amino acid esters and allenylic alcohol esters. This reaction, which allows efficient access to the key reported intermediate in the total synthesis of manzacidin C, uses a combined catalytic system of a chiral aldehyde, a palladium catalyst, and a Lewis acid. A series of α,α-disubstituted amino acid derivatives were generated with good-to-excellent experimental outcomes. Based on this methodology, the formal synthesis of manzacidin C and the stereodivergent synthesis of the chiral tetrahydropyrimidine-2-one core structures are achieved with high efficiency.

Scheme 1. Determination of the Absolute Configuration of Product 3a



■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.3c03762.

Experimental procedures, analytical data, and NMR and HPLC spectra (PDF)

Accession Codes

CCDC 2297525 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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